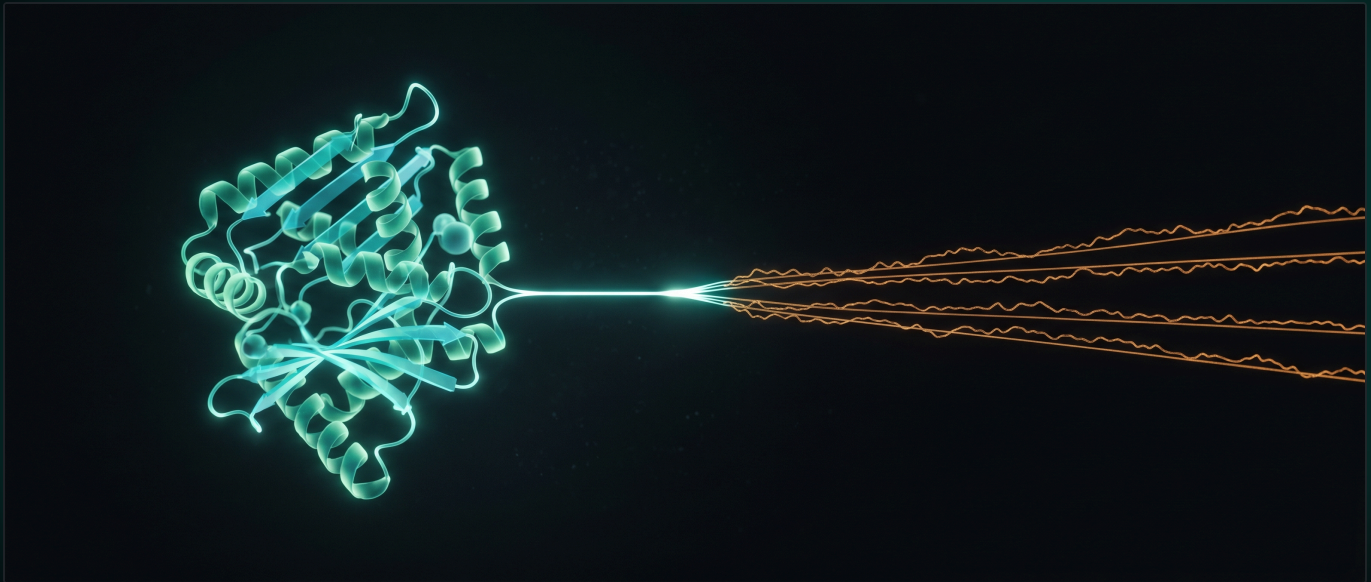




Judge Prep

Vinh Le

ENGINE C • DIGITAL PRODUCT PASSPORT • DATA ETL • LIVE ENDPOINT • DEMO
VIDEO

You own the most real thing in the room: a deployed compliance service on real public data, with a cryptographically verifiable passport. Lean into that. Everything here is traceable to the code or a live curl. Where the shipped scope is narrower than the spec, this doc tells you exactly how to say so, because the honesty is what wins.

— 01 THE THIRTY-SECOND VERSION

REWEAR-FUSED is **one computational co-design loop** that designs two matched molecules at once to close the textile-to-textile recycling loop on children's stretch apparel, which is non-recyclable today because of its elastane.

Instead of attacking existing elastane (crystalline, hostile, it defeats every native enzyme on the intact fiber), we invert the problem. **Engine A** screens a new, enzymatically-**cleavable** elastane fiber. **Engine B** designs the **matched** carbamate hydrolase built to cut that exact bond. **Engine C**, your lane, is the live Digital Recyclability Passport that proves a garment is safe and routes it.

"Carter's, here's the molecule for your next onesie, and the matched enzyme that closes its loop. No one has co-designed the pair, together, for textiles."

```
GARMENT → ENGINE C passport: clear / lab-test / divert ← your lane
           ↳ ENGINE A cleavable elastane (constraint-guided screening)
               ↳ [carbamate substrate target] → ENGINE B matched hydrolase
                   ↳ cleaves bond → recoverable monomers → NEW onesie
```

— 02 THE TWO WALLS (WHY THIS EXISTS)

Carter's collects worn baby clothes through **KIDCYCLE / TerraCycle**, then shreds them into pet bedding. Two walls stop a onesie becoming a onesie again:

- **Wall 1, chemical safety.** Nobody can prove the recovered fabric is chemically safe for infant skin. **This is the wall Engine C knocks down.**
- **Wall 2, elastane.** Nobody can break down the elastane in every stretch garment. (Engines A + B.)

LEAD WITH THIS

Elastane is in **~80%** of all clothing, and as little as **~1%** of it makes a recycler reject the **whole** garment. (Fashion for Good, 2026.)

SCALE

~92M tonnes of textile waste a year (UNEP); only **~1%** recycled fiber-to-fiber today.

03 WHAT IS REAL, WHAT IS IN-SILICO (THE HONESTY LEDGER)

Memorize the tiers. **Your engine is the top of this table**, the one component nobody can call a concept.

COMPONENT	TIER	TRL	WHAT IT IS
Engine C, the Passport	WIRED LIVE	6	real FastAPI endpoint + deployed UI; the Year-1 deployable product
The app (5 views)	WIRED LIVE	-	on Vercel, reading the signed artifact bundle through real code
Credential verify + QR	REAL	-	real WebCrypto Ed25519 in-browser; the QR is a working GS1 resolver
Training data	REAL DATA	-	631 real CPSC recalls, wired today; EU/PFAS/OEKO-TEX roadmap
Engine A, the fiber	IN-SILICO	3	constraint-guided screening; benchmark-referenced predictions
Engine B, the enzyme	IN-SILICO	2	LigandMPNN redesign (TM ~0.8); de-novo fold is the deferred swing
Wet-lab validation	FORWARD	-	the Cox Cleantech Residency ask

04 THE DE-NOVO QUESTION (THE ONE TRAP TO NEVER FALL IN)

HONESTY GUARDRAIL · EVERYONE SAYS THIS THE SAME WAY

The shipped enzymes are **LigandMPNN redesigns** of real serine-hydrolase scaffolds (TM 0.79 to 0.85), **not de-novo folds**. The de-novo fold (TM < 0.5, no prior art) is the GPU run we **deferred**. The "shouldn't be possible" novelty is the **co-design loop itself**, which is true and runs live. (This is Pravin's lane; you just need to never call the enzyme de-novo.)

05 THE FIVE SCIENTIFIC CONSTRAINTS (QUICK REFERENCE)

CONSTRAINT

- 1 **Aliphatic isocyanates only** (HDI/IPDI/H12MDI). Your passport surfaces this as "aromatic amine release: NONE".
- 2 **Ser-His-Asp triad, near-neutral pH 7.0-8.5** (cotton-compatible; not the wild-type pH ~10).
- 3 **Cleavable bond in the soft segment; hard segment 25-35 wt%** (keeps >400% stretch).
- 4 **Constraint-guided screening, not generative** (Engine A enumerates, scores, filters).
- 5 **In-silico, benchmark-referenced** (the molecular side; Engine C is the live exception at TRL 6).

— 06 THE EIGHT JUDGE-DEFENSE ANSWERS (UNIVERSAL)

Q What toxic products come off when you cleave it?

A None of concern. Aliphatic isocyanates only, so cleavage gives aliphatic diamines, not aromatic MDA/TDA carcinogens. The passport shows a green "aromatic amine release: NONE".

Q Doesn't your enzyme need pH 10, which destroys the cotton?

A That's the wild-type trap. We design a Ser-His-Asp triad that works near-neutral, pH 7.0-8.5, cotton-compatible.

Q Won't making it cleavable destroy the stretch?

A No. The cleavable bond lives in the soft amorphous segment, not the load-bearing crystalline domains. 25-35 wt% window, 440-620% elongation.

Q Can AI really invent a new block copolymer?

A We never claim that. Engine A is constraint-guided screening: enumerate 612, score 487, filter 23, surface 10.

Q Did you actually make any of this in a lab?

A No, entirely in-silico for the molecules (Engine A TRL 3, Engine B TRL 2). Engine C, my lane, is TRL 6 and runs live.

Q What's the business model?

A We replace the TerraCycle downcycle line item with a fiber-to-fiber loop plus an SB 707 / EU-DPP compliance asset. We're the Nexus Circular of textiles.

Q Isn't this two projects?

A One inverse-design loop. The carbamate bond in the fiber is the exact bond the enzyme cuts; topPairs.json cross-references each enzyme to its matched fiber.

Q Will Carter's pay for this?

A They already pay TerraCycle to downcycle. Engine C lowers their SB 707 eco-modulated EPR fees and pre-positions EU DPP. Same spend, better outcome, deployable in Year 1.

— 07 DO NOT SAY (UNIVERSAL)

- ✗ Not "de-novo enzyme" for the shipped designs (redesigns, TM ~0.8). Not "72 hours" / build-duration (DQ risk).
- ✗ Not "we spun a fiber" / "expressed the protein". Not "generative polymer design". No invented kinetics.
- ✗ Not FIFA tickets "won". No DOI outside the verified section-9 list.

"We're the Nexus Circular of textiles." Cox Cleantech led a \$150M round to majority-own Nexus Circular. The ask: \$10K + 6 weeks to validate the top pair, file a provisional, pilot one Carter's SKU. **Engine C is the part Carter's deploys first.**

A hybrid neuro-symbolic compliance system. A **deterministic regulatory rule layer runs first**; a **CatBoost text classifier** colors the residual rare-hazard categories. It routes each garment clear / lab-test / divert and emits a cryptographically signed Digital Product Passport. It is deployed, live, and trained on real public data.

```
GARMENT → [1] deterministic rules (cite the statute)
           FAIL → divert    FLAG → lab-test    else → clear
           → [2] CatBoost text model colors rare residual hazards (>0.5 → lab-test)
           → [3] sign W3C VC 2.0 passport (Ed25519) + GS1 Digital Link QR
served by FastAPI on Render; frozen-parity self-check refuses to boot on drift
```

The real data (say the number with confidence)

- **631 unique real CPSC SaferProducts recalls**, pulled live from `saferproducts.gov/RestWebServices/Recall` across 15 apparel/children-textile query terms, deduped by RecallID, saved to `cpsc_recalls.parquet`. Corpus spans 1974 to 2026-06-11. **No synthetic rows** (we explicitly rejected the spec's synthetic-SKU suggestion, the APEX lesson).
- **3 trainable labels** with real positive counts: flammability/sleepwear **n=206**, drawstring **n=142**, lead substrate **n=12**.

The model + the metric honesty

- **CatBoost** text classifier, version `catboost-text-v1.0` (iterations 350, depth 4, lr 0.08, Logloss), out-of-fold 5-fold StratifiedKFold.
- **We report PR-AUC, not ROC-AUC**, because positives are a minority (<5% per label; lead has only 12). ROC is misleadingly high on rare positives. Frozen macro: **PR-AUC 0.732**, ROC-AUC **0.951**.
- **Anti-leakage**: we strip the label-defining keywords (lead, phthalate, flammable, sleepwear, drawstring, pfas) from the feature text before training, so the model must predict from genuine correlates, not relearn the labeling keyword and report a meaningless ~1.0.

— 10 ROUTING + THE THREE GARMENTS

Rules run first: any **FAIL** routes to divert; else any **FLAG** routes to lab-test; else clear. The 3 demo decisions are **rule-driven** (the ML metric is the hazard classifier's credibility number, not what decides these three).

GARMENT	COMPOSITION	DECISION	WHY
Cotton Bodysuit 0-3M	100% cotton	CLEAR 0.974	ESPR pass, amines pass
Cotton-Blend Legging 12M	95% cotton / 5% elastane	LAB-TEST 0.910	ESPR FLAG (blend, not fiber-to-fiber)
UPF Rashguard 24M	82% rec. poly / 18% elastane	DIVERT 0.975	ESPR FAIL (≥15% elastane) + PFAS FLAG (DWR)

— 11 THE PASSPORT (REAL CRYPTOGRAPHY)

- **W3C VC 2.0** JSON-LD, UNTP DPP context, type [VerifiableCredential, DigitalProductPassport].
- **Real Ed25519 signature** (proof type Ed25519Signature2020) from issuer did:key:z6MkuBjwGH...V966Uv (a spec-compliant did:key, seed-derived so the demo reproduces). Stephen's frontend re-verifies it in-browser with a tamper toggle.
- **GS1 Digital Link**. The credential carries the canonical id.gs1.org/01/{GTIN-14}/21/{serial} (valid GS1 mod-10 check digit); the scannable QR points at our deployed origin, where our resolver 307-redirects to the live passport on a phone. (Have judges scan, not type the canonical string.)

— 12 LIVE ENDPOINT + FROZEN-MODEL PARITY

VERIFIED LIVE TODAY

```
rewear-engine-c.onrender.com/healthz → status ok, version 1.0.0, parity_kind output,
output_sha_ok TRUE, frozen_parity_sha bb95f4e8...5
routes: /healthz /garments /regulations /classify/{id} /passport/{id}.
Live /classify/carters-rashguard-swim returns divert 0.975, byte-matching the frozen artifact.
```

- **Frozen-parity gate:** `_verify_frozen_parity()` runs at import, hashes every file in the manifest, and **refuses to boot** (uvicorn never binds) on any SHA mismatch. Proven by refuse-to-boot tests.
- **It is OUTPUT parity, not model-binary parity** (be precise here, see do-not-say). The served classifications/passports are byte-identical to the frozen bundle; the field is `output_sha_ok`. We do not hash the .cbm binary because CatBoost bytes are not reproducible across Python 3.12.10 (build) vs 3.12.5 (Render), which would brick the boot on a benign diff.
- **Slim runtime:** the Render service installs only fastapi + uvicorn; the heavy ML/chem stack is build-only, so the free plan can't OOM on wheels it never loads.
- **Keepalive cron** (* / 10 * * * *) curls /healthz to keep the free-tier service warm. **Still pre-warm before judging: cold-start is ~21s, warm is 0.16s.**

— 13 YOUR HONESTY GUARDRAILS (ENGINE C SPECIFICS)

- ✗ **Not "85+ thresholds enforced"**. Three deterministic rules are wired today (ESPR elastane, AB 1817 PFAS, REACH Entry 43 amines) over a 7-regulation catalog. The README says "~20% of the spec / reference implementation". 85+ is the roadmap target.
- ✗ **Not "trained on CPSC + EU Safety Gate + PFAS + OEKO-TEX"**. Only CPSC (631 recalls) is wired. DevPost lists the others; reconcile to *"CPSC is the wired source today; EU Safety Gate, the Peaslee PFAS set, and OEKO-TEX negatives are named in our roadmap."*
- ✗ **Not "PR-AUC 0.85"**. That is the handoff target. The real frozen macro is **0.732**. Report the achieved number.
- ✗ **compositionDrivers is NOT SHAP**. They are real composition ratios (elastane %, synthetic %, archetype flag), deliberately not dressed as model explanations. SHAP is roadmap.
- ✗ **The Render API's /judges route 404s** on the live deploy (it predates that commit). Don't promise "hit /judges on the API". `recallCount=631` is verifiable via the manifest + `/healthz`. (The frontend /judges page on Vercel is fine.)
- ✗ **Parity is OUTPUT, not model-binary**. Say "output parity, byte-identical served outputs", not "the model binary is hash-locked".

— 14 YOUR Q&A

Q Is this trained on real data or did you fake the dataset?

A Real. 631 unique CPSC SaferProducts recalls pulled live from saferproducts.gov, saved to a parquet, count verified. No synthetic rows. The spec suggested a synthetic 10,000-SKU population; we explicitly rejected it to avoid the synthetic-data failure mode.

Q Why PR-AUC instead of accuracy or ROC-AUC?

A Positives are a minority (<5% per label; lead has 12 in 631). ROC is misleadingly high there, so we report PR-AUC (average precision) out-of-fold via 5-fold CV. Macro 0.732 / ROC 0.951. The slow test asserts flammability PR-AUC > 0.7 and lead PR-AUC < 0.8, since lead is rare and honestly hard from text.

Q How do I know the model didn't memorize the labeling keyword?

A `model.py` strips the label-defining terms from the feature text before training. With them gone the model must predict the hazard from product type, brand, and co-occurring language. Without the strip it would relearn the keyword and report a meaningless ~1.0.

Q Is the live endpoint serving the same model that made your artifacts?

A Yes, enforced by hash. `_verify_frozen_parity()` hashes every manifest file at import and refuses to boot on mismatch. `/healthz` returns `frozen_parity_sha bb95f4e8` and `output_sha_ok true`; `/classify` byte-matches the artifact. It is output parity by design, not a model-binary hash.

Q Is the passport signature real or a placeholder?

A Real Ed25519. `build.py` signs the canonical JSON of the credential with an Ed25519 private key, producing the base64 `signatureValue`. The issuer is a spec-compliant `did:key`. The frontend verifies it in-browser, no server, with a live tamper toggle.

Q How many regulations does it really enforce?

A Three deterministic rules wired today over a 7-regulation catalog, each citing its real statute. The full target is 85+ thresholds; the README labels the shipped state ~20% of the spec. What is wired is real and cited; the rest is roadmap.

Q Will the demo break if Render is down?

A No. The frontend renders the passport view entirely off the static signed artifacts. The live endpoint proves it is a real deployed service; the static bundle guarantees the demo runs offline. A keepalive cron keeps the free-tier service warm.

15 THE DEMO VIDEO

You own the DevPost cut: a ~2:30-3:00 hand-edited video, real screen capture of the deployed app off the verbatim `demo_script.md`. Live link verified public: youtube.com/watch?v=TP05nhIIWrI. Before the deadline locks: confirm it plays in incognito, captions on, Public + "Not for Kids".

REWEAR-FUSED JUDGE
PREP

VINH • ENGINE C • HAND TO STEPHEN (FRONTEND) OR PRAVIN (MOLECULES) OUTSIDE YOUR
LANE